

SIH 26188 — AI-Based Fake Identity & Document Screening System

Technical Specification, System Architecture & Dataset Inventory

Problem Statement ID: 26188 **Organisation:** Ministry of Home Affairs **Department:** Sashastra Seema Bal (SSB), Police II Division **Category:** Software

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1. Problem Statement Summary

Operational context

Border checkpoints process thousands of identity documents daily — passports, visas, national ID cards, permits, travel authorisations. Manual verification is slow, error-prone, and cannot reliably detect sophisticated forgeries.

Threats to be addressed:

Fake passports and visas

Altered photographs

Modified dates of birth

Tampered visa stamps

Identity impersonation
Multiple identities used by the same person
Expired or blacklisted travel documents
High passenger volume causing delays

Strategic note on SSB's mandate

Sashastra Seema Bal is the border guarding force for the **India–Nepal (1,751 km)** and **India–Bhutan (699 km)** borders. These are *open borders* — Nepali and Bhutanese citizens do not require a passport or visa to enter India under the respective friendship treaties.

This changes the document mix in the field significantly:

Document	Relevance to SSB
Nepali Citizenship Certificate	Very high — primary ID at Indo-Nepal crossings
Indian Aadhaar / Voter ID / Passport	Very high — Indian nationals crossing
Nepali passport	High
Bhutanese permits and Voter ID	Moderate
Third-country passports + visas	Lower volume, higher risk

Recommendation: build the international passport/visa pipeline (it demonstrates ICAO-standard rigour), but make the *demo narrative* about the documents SSB actually inspects. Explicitly acknowledging the open-border reality signals domain understanding that most competing teams will not have.

Required modules

Module	Objective
1 — OCR Extraction	Extract all relevant fields from identity documents
2 — Document Validation	Verify extracted information against official standards
3 — Tampering Detection	Detect digitally or physically altered documents
4 — Face Verification	Confirm document owner matches presented individual
+ Risk Score	Fuse all signals into an actionable decision for the officer

Expected impact

Reduce verification time from minutes to seconds

Improve forged/tampered document detection
Standardise screening decisions across checkpoints
Enable data-driven risk assessment
Create a digital trail for investigation and intelligence

2. Critical Assessment

These are the errors most commonly made on this problem statement. Fixing them early saves weeks.

2.1 Metadata analysis is near-useless at a physical checkpoint

If the officer scans the document at the counter, **the scanner writes the file**. EXIF will always look clean, timestamps will be consistent, and the image will be single-JPEG. A professionally printed physical forgery passes ELA, double-JPEG detection, and EXIF checks *completely clean* — because the image genuinely is a fresh scan. It is the physical object that is fake.

Tampering detection therefore splits into two tracks that share almost no code:

Track	Applies to	Techniques
Digital file forensics	Pre-submitted files — e-visa uploads, emailed scans, online applications	ELA, double-JPEG/DCT, EXIF, copy-move, splice segmentation
Physical artifact forensics	Live scans at the counter	OCR-B font conformance, layout template geometry, guilloche continuity, print halftone analysis, hologram/OVD presence, ghost portrait check, stamp geometry

Most teams build ~90% digital. The SSB use case is ~90% physical. **Build the physical track as primary.**

2.2 PRNU is the wrong term and the wrong technique

PRNU (Photo Response Non-Uniformity) is sensor-noise fingerprinting. It requires ~50+ images from a *known* camera to estimate a reference pattern. You have one image from an unknown device.

What you can actually do is **noise residual inconsistency**: apply SRM/high-pass filters, then look for regions whose residual statistics differ from their surroundings. Related concept, different method. Do not say "PRNU" in your pitch.

2.3 ELA is weaker than commonly claimed

Error Level Analysis is useful as a *visualisation* for the officer, but as a signal it is fragile:

Global recompression after tampering flattens the difference
Mixed-quality source images produce false bright patches

PNG and PDF input yield nothing at all

Keep it, weight it low, and never let it carry a forgery claim alone. State the limitation before a judge does.

2.4 A per-country stamp template library is not achievable

Thousands of entry/exit stamp variants exist across countries and ports, with no public reference set.

Replace template matching with logic that needs no templates:

Duplicate stamp detection — two stamps matching at pixel level are copy-pasted. Strongest and cheapest stamp signal.

Date-logic consistency — stamp date within visa validity, entry before exit, stay duration within permitted duration.

Count vs declaration — number of entry stamps vs declared trips.

Placement validity — stamp on a page that should not carry one.

2.5 The synthetic-training trap

If you generate synthetic splices and train a CNN on them, it learns *your splice generator's artifacts*, not forgery. You will score 99% on your held-out set and near-chance on real forgeries.

Mitigation: train on one tampering method family, test on a *different* one. Train on SIDTD (crop-and-replace + inpainting), test on IDNet or MIDV-Holo. Report both numbers. This is more credible than a single high number.

2.6 The face match score rescaling is misleading

The common formula $\left(\frac{\cos + 1}{2} \right) * 100$ maps:

Cosine similarity	Reality	Displayed score
0.00	Total stranger	50
0.16	Confident impostor	58
0.60	Strong genuine match	80

An officer reads "58%" as "more than half, probably him." The scale is wrong in the direction that admits fraudsters.

Fix: either fit a logistic calibration on your genuine/impostor score distributions so the number is an actual probability, or drop the percentage and show a three-band verdict plus the margin from threshold ($\frac{\cos - 0.52}{\text{threshold } 0.40, +0.12 \text{ above}}$).

2.7 Document-photo-to-live is a different domain from live-to-live

Passport photos are printed, halftone-screened, overprinted with guilloche and holograms, sub-300 DPI once scanned, and often up to ten years old. ArcFace degrades measurably.

Calibrate a separate threshold on doc-vs-live pairs. It will be lower than the standard live-to-live figure. Using 0.4 unmodified produces an FRR that makes the system unusable.

2.8 Redundant detection wastes latency

A common pattern runs `RetinaFace.detect_faces()` then `FaceAnalysis.get()` — which internally runs RetinaFace *again*, plus alignment, plus embedding. That is two full detector passes per face, four per comparison.

Fix: Use `FaceAnalysis` once and read `.embedding` off the result. Set `det_size=(320, 320)` — the `buffalo_l` default of 640×640 is unnecessary for a cropped ID photo.

2.9 Liveness is lower priority than it appears

At a manned SSB checkpoint the officer is physically looking at the person. A printed-photo attack is not a realistic threat there. Liveness matters for e-gates and remote e-visa enrollment.

Build it — it demos well and future-proofs the e-gate story — but do not let it consume time needed for MRZ and physical-forgery work. Scoping this explicitly reads as operational understanding.

2.10 Missing sub-module: 1:N duplicate identity search

"Multiple identities used by the same person" is **not** 1:1 verification. It is 1:N search: embed the live face, query a gallery of prior enrollments, flag near-duplicates carrying different document numbers.

Different code path, different index, different threshold. Most teams miss this. It is explicitly named in the problem statement.

2.11 FAR/FRR is an operator decision, not a constant

"Tune for very low FAR" is not a plan. At 5,000 passengers/day, an FRR of 2% is 100 secondary inspections daily.

Expose the FAR/FRR curve in the UI and let the checkpoint commander set the operating point. Far stronger than a hardcoded threshold.

2.12 Roboflow is a build-time tool, not a runtime dependency

Border checkpoints have no reliable internet, and hosted inference adds 200–800 ms per call. Both violate the core requirements.

Use Roboflow to annotate, version, and train. Then export weights and run them locally — ONNX in your own runtime, or Roboflow Inference in a Docker container on the same box. At demo time nothing leaves the machine. "Fully air-gapped operation" is a differentiator with MHA.

Assumption	Reality
"OCR is available on Roboflow"	Roboflow does detection, segmentation, classification. It draws boxes; it does not read characters. You still need PaddleOCR/docTR, plus the parser, check-digit logic, and field normalisation. Roboflow covers ~25% of Module 1
"Face detection is available on Roboflow"	InsightFace <code>buffalo_l</code> bundles RetinaFace detection + 5-point alignment + ArcFace embedding, optimised. A community Roboflow face detector gives worse boxes, no landmarks, and an extra model in your latency budget. Roboflow's only genuine Module 4 contribution is the anti-spoofing classifier
"Tampering must be built by us"	Correct. But Roboflow is still the right <i>tool</i> for building it — annotation, versioning, training on your synthetic forgery set

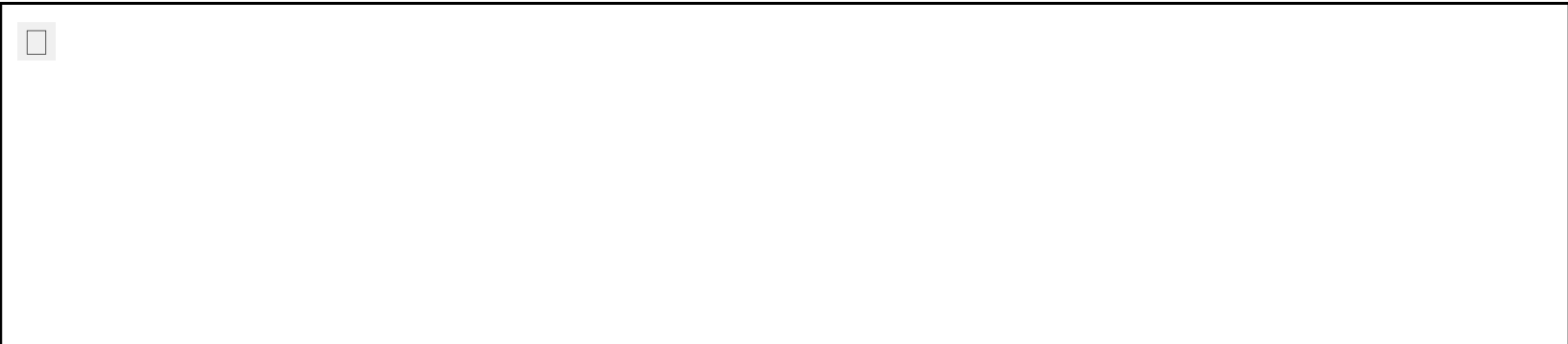
3. System Architecture

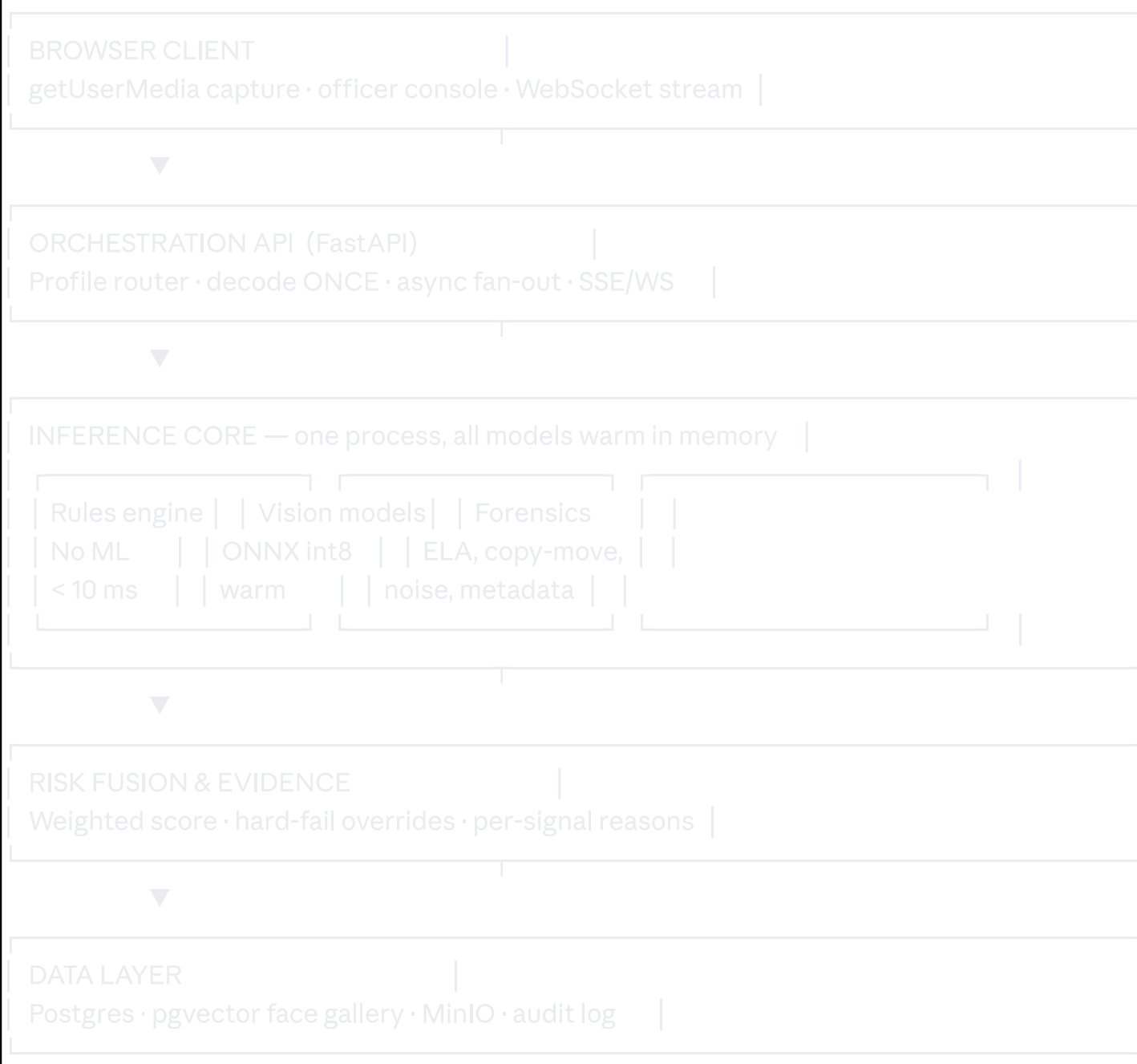
3.1 Three separate architectural decisions

The "module-wise vs document-wise" question conflates three things. The answer differs for each:

Axis	Decision	Rationale
Code organisation	Module-wise	Each module is a package with one clean entry point returning typed signals. Testable, parallelisable across teammates
Execution routing	Document-wise	A YAML profile per document type decides which checks run and how they are weighted
Deployment	Single process	This is the latency answer — see 3.3

3.2 Runtime layers





3.3 Three properties that matter more than the boxes

Decode once. The orchestration layer decodes the upload into a single numpy array and passes it *by reference* to every module. No module re-reads bytes from disk or receives an image over HTTP.

This is why the inference core is one process, not four microservices. Microservices per module means each gets the image via encode → HTTP transfer → decode → infer → encode result. A 3 MB scan serialised four times costs more wall-clock than the inference itself.

Rules engine runs first and can short-circuit. MRZ checksum failure, expired document, or watchlist hit returns RED before any model is touched. That path is under 200 ms with no CV at all.

Results stream as they land. The client holds a WebSocket open:

The officer reads the first result while the third is still computing. Perceived latency is what you are actually optimising.

3.4 Document profiles (execution routing)

```
❏
yaml
# profiles/passport.yaml
passport:
  checks: [mrz, viz_fields, face_doc, stamps, ela, copy_move, ocrb_font]
  weights:
    mrz_checksum: 0.30
    viz_mrz_mismatch: 0.35
    face_match: 0.20
    tamper_heatmap: 0.10
    stamp_duplicate: 0.05
  hard_fail: [mrz_composite_checksum, expired, watchlist_hit]

# profiles/aadhaar.yaml
aadhaar:
  checks: [qr_verify, viz_fields, face_doc, hologram, guilloche]
  weights:
    qr_signature: 0.50 # cryptographic — dominates
    qr_viz_mismatch: 0.25
    face_match: 0.20
    hologram_present: 0.05
  hard_fail: [qr_signature_invalid, verhoeff_checksum_fail]
```

Aadhaar's signed QR is a **cryptographic** integrity check — stronger than passport MRZ check digits. See section 4.2.

3.5 Latency budget (CPU-only, ~1.4 s p95)

Stage	Budget
Decode, deskew, quality gate	80 ms
Doc type + segmentation crop (yolov11n-seg @ 640)	40 ms
MRZ detect + OCR the strip only	120 ms
VIZ field boxes + OCR crops only	250 ms
Face detect + embed, both images	200 ms
ELA + copy-move + stamp checks	150 ms
Fusion	< 5 ms

Five things that actually move the number:

Never OCR the whole page. Detect field regions, OCR only those crops. 5-10× win. This is why the field-detection model matters more than the OCR engine choice.

Gate on cheap deterministic checks. MRZ checksum, expiry, watchlist hash — all under 200 ms, no models. Hard-fail returns RED without touching the CV stack.

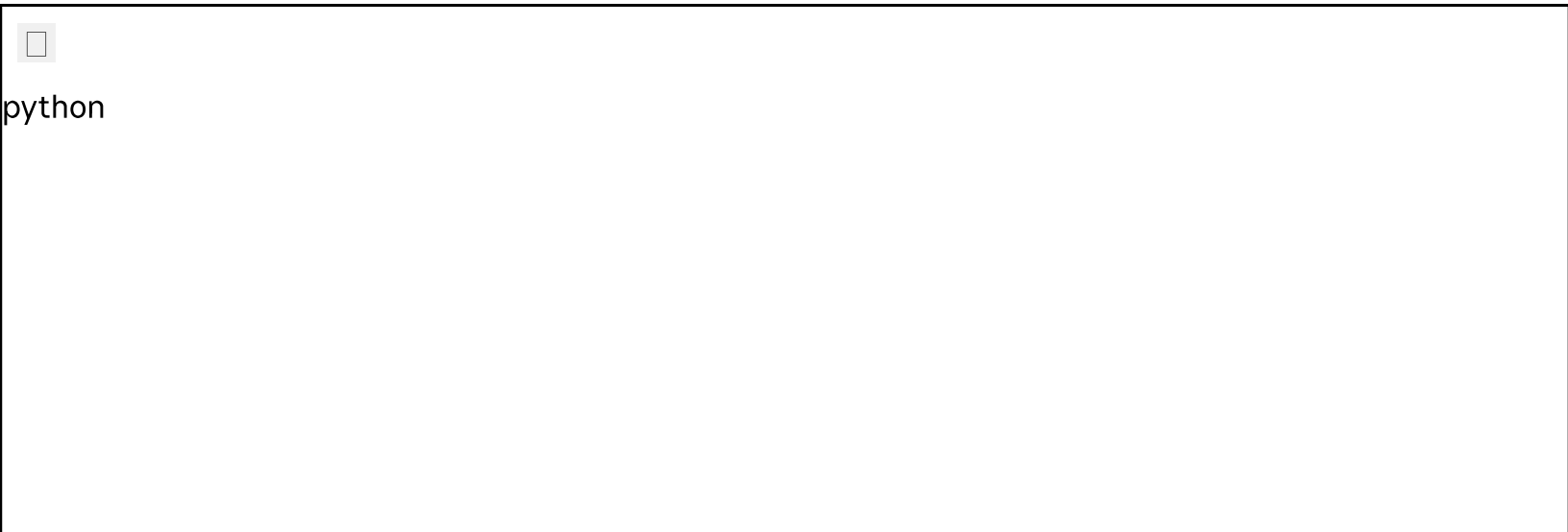
Stream over WebSocket as each module finishes. Nearly free, transforms perceived latency.

ONNX Runtime, int8 quantised, `intra_op_num_threads` tuned to the box, sessions created once at startup — never per-request.

Stay CPU-only and say so. Border posts do not have GPUs. "Runs on a 4-core box with no internet" beats "0.3 s on an A100" with this audience.

3.6 The signal contract

Freeze this in week one. It is what lets four people work in parallel without integration hell.



```
from dataclasses import dataclass
```

```
@dataclass
```

```
class Signal:
```

```
    id: str          # "mrz.checksum.dob"
```

```
    module: str       # "validation" | "tamper" | "face" | "extraction"
```

```
    verdict: str      # "pass" | "fail" | "inconclusive"
```

```
    confidence: float  # 0.0 - 1.0
```

```
    weight: float      # from the document profile
```

```
    hard_fail: bool    # bypasses the weighted score entirely
```

```
    evidence: str      # "MRZ DOB 1991-03-04 != printed 1991-08-04"
```

```
    region: tuple | None # bbox for the UI overlay
```

```
    latency_ms: int
```

Every module returns `list[Signal]`. Fusion consumes only this. No module imports another module's internals.

4. Module Specifications

4.1 Module 1 — OCR Extraction

Pipeline: capture → quality gate → segmentation crop → perspective correct → doc type classify → field detect → OCR crops → parse → normalise

The MRZ is the highest-value component. Every ICAO-compliant passport carries a machine-readable zone: two lines of 44 characters encoding name, document number, nationality, DOB, sex, and expiry — each with **check digits**, plus a composite check digit over the whole thing.

This gives you two things nearly free:

Self-validating data. A forged document number that fails its own check digit is caught by arithmetic. No ML.

A cross-check axis. Amateur forgers edit the printed date of birth (the Visual Inspection Zone) and forget the MRZ, or vice versa. **VIZ ↔ MRZ mismatch is your most reliable, most explainable tamper signal** — and it demos beautifully, because you can show exactly which two fields disagree.

Build MRZ parsing in week one. It makes Modules 1 and 2 largely the same code and gives Module 3 a strong non-ML signal to lean on.

Good news on Indian passports: they use standard ICAO TD3 format. All MRZ work on MIDV-2020's European passports transfers directly — same two 44-character lines, same check-digit algorithm, same OCR-B font. Only the VIZ layout differs, which is a field-detection retrain, not new science.

OCR configuration:

Engine: PaddleOCR or docTR (better on ID-card layouts than Tesseract; both have ONNX exports)
MRZ strip: restrict the charset to OCR-B alphanumerics + <. Massive accuracy gain.
VIZ fields: OCR crops only, never the full page

Extracted fields:

Passport	Visa	Aadhaar	PAN
Name	Visa Number	Name	Name
Passport Number	Visa Type	Aadhaar (masked)	Father's Name
Nationality	Entry Validation	DOB / YOB	DOB
Date of Birth	Stay Duration	Gender	PAN Number
Date of Expiry	Validity dates	Address	
Gender	Passport Number ref VID		

4.2 Module 2 — Document Validation

Six layers. **The first four require zero ML, zero network, and run in under 10 ms combined.** They are the backbone and the fast path.

Layer A — Intra-document integrity (cryptographic or arithmetic)

Document	Check
Passport	Five MRZ check digits: document number, DOB, expiry, optional data, composite
Aadhaar	Verify the UIDAI-signed Secure QR. Also Verhoeff checksum on the 12-digit number
PAN	Format + structural rules
Driving Licence	State code + RTO code validity

The Aadhaar QR is your single strongest asset for Indian documents.

The Secure QR contains the last 4 digits of the Aadhaar number, name, address, gender, date of birth, and photograph — signed with UIDAI's digital signature, making it tamper-proof. The public certificate for signature validation is downloadable from UIDAI.

This means you can verify an Aadhaar card **cryptographically, offline, with certainty** — not a confidence score, a signature that validates or does not. If someone photoshops the DOB, the printed text no longer matches the signed payload, and the QR cannot be forged without UIDAI's private key.

Validation flow: verify QR signature → compare signed payload against OCR'd printed text → any mismatch is a hard fail with cryptographic proof. This requires **no training data at all**.

Layer B — Format and schema conformance

Field lengths and character sets per ICAO Doc 9303

Nationality against ISO 3166-1 alpha-3 (note MRZ quirks: D for Germany, GBR subtypes)

Date formats, gender codes (M / F / <), document type codes

Layer C — Cross-field logic within one document

DOB < issue date < expiry date

Expiry – issue equals a standard validity period (India: 10 years adult, 5 years minor). An 8-year window is fabricated.

Age at issue consistent with the validity granted

Expiry not in the past

Layer D — Cross-document consistency (*most teams miss this*)

Name on visa matches passport (after transliteration normalisation)

Passport number printed on the visa matches the passport's own number

Visa validity cannot extend beyond passport expiry — a visa outliving its passport is a hard fail

Stamp dates fall inside visa validity

Layer E — External lookups (mocked for demo)

Watchlist / blacklist by document number and by name + DOB

Lost and stolen document registry (Interpol SLTD is the real system; seed a local table)

Prior-crossing history for this document number

Layer F — Temporal and behavioural

Same document number scanned at two checkpoints too far apart in too little time

Exit without matching entry, or duplicate entries

Free, because you already have the audit log

Layers A-D are deterministic, explainable, instant — the backbone. **Layers E-F** turn this from a screening tool into an intelligence tool, which is the language the problem statement uses.

4.3 Module 3 — Tampering Detection

Two tracks (see 2.1). Present this as a **signal ensemble** where each signal is independently explainable — never as "we trained a model that detects fakes."

Track A — Physical artifact forensics (PRIMARY for SSB)

Signal	Method
OCR-B font conformance	Character stroke width, glyph geometry in the MRZ strip vs the ICAO spec
Layout template geometry	Field positions relative to document corners vs the official template
Guilloche continuity	Security background pattern must be continuous under text and around the photo
Ghost portrait	Present, correctly positioned, and matching the main photo
Hologram / OVD	Presence and chromaticity behaviour (multi-angle capture)
Print halftone analysis	Genuine printing has a specific screen pattern; inkjet forgeries do not
Photo boundary	Physical edge artifacts where a photo was cut and glued
Stamp logic	Duplicate detection, date consistency, count vs declaration

Track B — Digital file forensics (SECONDARY, for e-visa flow)

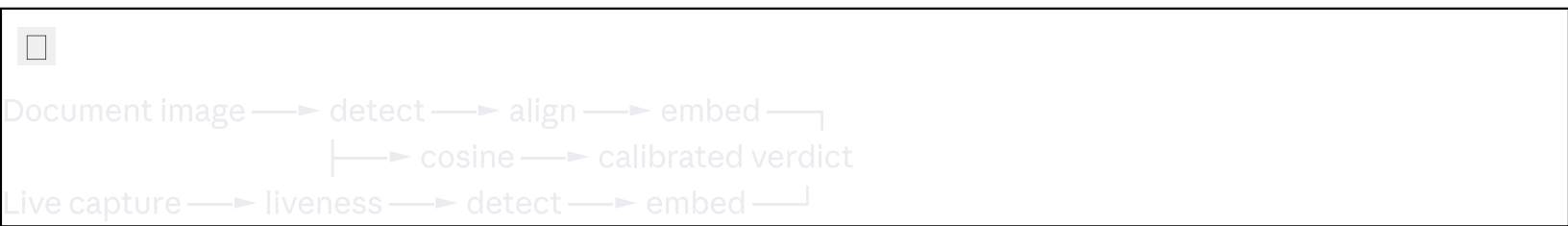
Signal	Method	Weight
Error Level Analysis	Recompress at known quality, diff. Good for the officer overlay	Low
Noise residual inconsistency	SRM/high-pass filter, regional statistic outliers	Medium
Copy-move detection	SIFT/ORB keypoint matching within one image	High
Double-JPEG / DCT	Compression history analysis	Medium
Metadata	<code>exiftool</code> / <code>pikepdf</code> . <code>Software: Adobe Photoshop</code> on a "scan" is a free win	High when present
Learned localisation	Segmentation model producing a heatmap, not a binary label	Medium

Explainability: use a **segmentation model outputting a mask**, not Grad-CAM on a classifier. Grad-CAM localisation is weak and hard to defend under questioning.

Build order: classical signals first (they always work, no data needed), learned model second (it might not).

4.4 Module 4 — Face Verification

4a — 1:1 Verification



Component	Choice
Detection + alignment + embedding	InsightFace <code>buffalo_1</code> , single pass, <code>det_size=(320,320)</code>
Embedding	ArcFace, 512-d
Metric	Cosine similarity
Threshold	Calibrated separately on doc-vs-live pairs — not the live-to-live default
Output	Three bands (MATCH / MANUAL_REVIEW / MISMATCH) + margin from threshold

Quality gates before comparing (reject and prompt re-capture rather than compare badly):

Blur — Laplacian variance threshold
Brightness / contrast
Face size relative to frame
Yaw / pitch / roll angle

4b — 1:N Duplicate Identity Search

The problem statement's "multiple identities used by the same person." Embed the live face, query a pgvector index over all prior enrollments, flag near-duplicates carrying *different* document numbers. Separate threshold from 1:1 — 1:N needs a stricter one because you are running many comparisons.

At demo scale, `pgvector` with an HNSW index is plenty. Mention FAISS/Milvus for production scale; do not build it.

Bias evaluation is mandatory here. Face recognition has documented accuracy disparities across demographics (see NIST FRVT). In a border-security context with real consequences, validate across skin tones, ages, and genders using FairFace or RFW, and report the disparity. A panel that asks this question and gets a real answer will remember it.

Privacy: store embeddings + audit logs, not raw face images. Encrypt at rest. Do not retain live captures beyond the session unless legally mandated.

5. Risk Fusion Engine

Do not use gradient boosting. Two reasons: you have no real forgery distribution to train weights on, and a border officer detaining someone needs a *justification*, not a number.

Design



Hard-fail conditions bypass the score entirely and go straight to RED:

MRZ composite check-digit failure
Aadhaar QR signature invalid
Document expired
Watchlist hit
Visa validity exceeds passport validity

Every contributing signal surfaces as an evidence card:

MRZ DOB 1991-03-04 ≠ printed DOB 1991-08-04 — weight 0.35 — [highlight region]

Keep weights in the YAML profile so you can tune them live on stage.

Officer console requirements

The UI is where you win or lose. Required elements:

Side-by-side document view with tamper heatmap overlay
Extracted fields with per-field confidence and validation status
Face match pair, side by side, with the verdict band
Risk band with the **evidence list**, not just a number
FAR/FRR operating-point control
Audit trail: score, model versions, timestamp, officer ID

Human-in-the-loop. This system assists, never replaces, the officer's decision — especially at MANUAL_REVIEW and near-threshold scores. State this explicitly in the pitch.

6. Dataset Inventory

6.1 The honest framing

There is no public dataset of real forged passports, and there is no legitimate public dataset of real Indian government IDs. There never will be — it is PII plus a forgery manual, and for Aadhaar specifically it is

criminal under Section 29 of the Aadhaar Act 2016.

Two consequences:

The **MIDV family from Smart Engines** is the backbone of essentially every usable dataset below. Roughly **40% of what you need, you generate yourselves**. Budget for it as real work.

Frame this as a strength in your pitch:

"Real Indian ID datasets cannot legally exist, so we built a synthetic generator from official specimens with checksum-valid identity numbers. For Aadhaar specifically, we need no forgery training data at all — we verify UIDAI's cryptographic signature."

6.2 Tier 1 — Core ID document datasets (Modules 1, 2, 4)

Dataset	What it gives you	Size	Where
MIDV-2020	The backbone. 1000 video clips, 1000 scans, 1000 photos of 1000 unique mock documents, each with unique field values and artificially generated faces. Ground truth includes ideal text field values, document quadrangle position, and face position in every frame. Trains doc localisation, field detection, OCR evaluation, and face-crop pipeline	72,409 images	ftp://smartengines.com/midv-2020 l3i-share.univ-lr.fr/MIDV2020/midv2020.html
MIDV-500	50 document types — more type diversity, weaker annotation. Use for doc-type classification robustness	500 clips	ftp://smartengines.com/midv-500
MIDV-2019	MIDV-500 recaptured under low light and heavy projective distortion. Hardens your quality gate and deskew	—	ftp://smartengines.com/midv-2019
DocXPand-25k	Independent synthetic ID generator + 25k images, with separate field crops (photos, ghost images, barcodes, datamatrices). License: CC BY-NC-SA 4.0 — non-commercial, fine for SIH, must be flagged	25k	github.com/quicksign/docxpan

Download MIDV-2020 first. Everything else in your pipeline can be built against it.

Gap: MIDV-2020 covers ten European document types. **No Indian documents.** That gap is filled by Roboflow + your own generation.

6.3 Tier 2 — Forgery and tampering (Module 3)

Dataset	What it gives you	Where
SIDTD	The most directly on-target set available. Extends MIDV-2020 — originals treated as bona fide, forged versions generated by Crop & Replace and inpainting, across ten European nationalities (Albanian, Azerbaijani, Estonian, Finnish, Greek, Lithuanian, Russian, Serbian, Slovakian, Spanish). Ships download code, a generator for new forgeries, and baseline models	github.com/Oriolrt/SIDTD_Dataset tc11.cvc.uab.es/datasets/SIDTD_1
MIDV-Holo	Your physical-forgery track. 700 video clips of artificial ID documents with and without holograms under varied lighting, plus 400 clips of presentation attacks: copy without hologram, hand-drawn pseudo-hologram, photocopy, and physical photo-replacement. Ships binary hologram location masks	github.com/SmartEngines/midv-holo data: ftp://smartengines.com/midv-holo License: CC BY-SA 2.5
IDNet	Larger and deliberately harder. The authors show its fraud samples achieve higher SSIM against originals than SIDTD's — meaning stealthier, closer to untampered. Use as your hard test set, not training	arxiv.org/abs/2408.01690
DocTamper	Large-scale document <i>text</i> tampering with pixel masks. Mostly Chinese, but text splicing artifacts transfer. Best available for text-manipulation detection	github.com/qcf-568/DocTamper
FindIt	Real (not synthetic) forged French administrative documents from the ICPR 2018 fraud detection contest. Small but genuinely human-made	ICPR 2018 Find-it contest page
MIDV-DynAttack	Extends MIDV-Holo with static and dynamic hologram attacks, tripling attack samples	github.com/EPITAResearchLab/pouliquen.25.icdar

Critical protocol: train on SIDTD, **test on IDNet or MIDV-Holo**. Reporting a number on a harder, differently-generated set is the difference between a credible claim and an overfit one.

6.4 Tier 3 — Generic image forensics (pretraining)

Document forgery sets are too small to train from scratch. Pretrain on these, then fine-tune.

Dataset	Purpose	Where
CASIA v1 / v2	Splicing and copy-move, standard baseline	Widely mirrored on GitHub/Kaggle
CoMoFoD	Copy-move specifically, with post-processing variants	vc1.fer.hr/comofod
DEFACTO	~150k manipulations from MS-COCO. Largest, best for pretraining	Kaggle / IEEE DataPort
IMD2020	Real-world manipulated images found in the wild	staff.utia.cas.cz/novozada/db
Columbia Uncompressed	Uncompressed splicing — clean signal for validating noise-residual code	Columbia DVMM

6.5 Tier 4 — Face (Module 4)

Dataset	Purpose	Where
LFW	Standard 1:1 verification protocol. Sanity-check your threshold code	vis-www.cs.umass.edu/lfw
AgeDB-30	Age-variant verification. The one that matters most — a 10-year-old passport photo is your hardest genuine case	InsightFace eval packs
CFP-FP	Frontal-to-profile. Live capture is not always frontal	InsightFace eval packs
FairFace	Balanced across race, gender, age. For your bias audit	github.com/joojs/fairface
RFW	Racial Faces in the Wild — standard demographic-bias benchmark	whdeng.cn/RFW

Do not download MS1M or VGGFace2. You are not training a face model; ArcFace weights are pretrained. Downloading a training corpus you will not use is a common time sink.

The unavoidable gap: no public dataset of ID-document-photo ↔ live-selfie pairs exists. It is inherently PII. MIDV-2020 gives document faces but no matching live captures. **You must build this yourself** — see section 8.

6.6 Tier 5 — Anti-spoofing / liveness

Dataset	Notes	Where
CelebA-Spoof	625k images, 10k subjects, openly downloadable. The practical choice	github.com/ZhangYuanhan-AI/CelebA-Spoof
LCC-FASD	Small, fast to iterate on for a demo	Kaggle
DLC-2021	Document Liveness Challenge — screen recaptures and printed copies of documents. Relevant to "is this a photo of a screen showing a document"	Smart Engines
CASIA-FASD, Replay-Attack, OULU-NPU, SiW	Academic standards, but all require a signed EULA from the host university . Turnaround exceeds hackathon timelines. Skip	—

6.7 Official specimen sources (template base for generation)

Published by the issuing authorities. Legally clean, and the ground truth your generator needs.

Document	Source
Indian passport	PRADO — EU Council public register of authentic identity and travel documents: consilium.europa.eu/prado . Has Indian passport specimens with security features annotated. Also Passport Seva / MEA security-feature lists
Aadhaar letter + PVC card	UIDAI publishes sample card images in guidelines and press releases. The PVC card carries a hologram, ghost image, and guilloche pattern — all named, checkable security features
PAN card	Income Tax Department official card format
Voter ID (EPIC)	Election Commission of India specimen
Driving licence	Parivahan / MoRTH — the smart card DL format is standardised under the Central Motor Vehicles Rules, so there is a written spec, not just an image
Passport MRZ layout	ICAO Doc 9303, free PDF, all 13 parts: icao.int

What	Where
ICAO Doc 9303 (MRZ spec, check digits, field layouts)	icao.int — free PDF
ISO 3166-1 alpha-3 country codes	pycountry Python package
OFAC SDN list — real sanctions list, free CSV/XML, includes names, DOBs, and passport numbers	treasury.gov/ofac downloads
UN Security Council Consolidated List — free XML	scsanctions.un.org
UIDAI Secure QR spec + public certificate	uidai.gov.in/en/ecosystem/authentication-devices-documents/qr-code-reader.html

On the sanctions lists: instead of inventing a fake watchlist, load a real published one. Your blacklist lookup then has realistic name-matching problems — transliteration variants, aliases, partial DOBs — which is exactly the hard part, and exactly what will impress a panel.

7. Roboflow Universe — Verified Assessment

7.1 Reality check

Universe is community-uploaded. Most entries are small hobby datasets; several are outright broken (class names that are literally Roboflow's CSV export boilerplate). **Use Roboflow for the localisation layer** — "where is the MRZ, where are the fields, where are the stamps." Keep the **forgery judgment** in your own logic. Nothing on Universe will tell you a passport is fake.

7.2 Verification results on commonly cited links

Project	Claimed	Verified	Verdict
centurion-.../aadhaar-card-details-extraction-n4hyw	6.3k images	2,645 images , classes ["0", "1", "2", "3", "4"] — unnamed integers, annotation map what classes 0–4 mean label reads "car" (leftover)	Usable only after you manually
cutm-iwh4a/aadhaar-card-details	Separate dataset	2,645 images, identical classes	Same dataset re-uploaded. 231 downloads vs 28 — this is the

Project	Claimed	Verified	Verdict
vexil-infotech/voterid-card-details-extraction	Voter ID extraction	31 images , 7 classes, Public Domain, "trained model" available	more-used copy. Do not count it twice Schema is good (Address, Age, Father's_Name, Gender, Husband_Name, IssueDate, Name). 31 images is not trainable . Use the schema, ignore the model
passport-stamp/passport-stamp	Entry/departure stamps	47 images , 3 classes (entry_stamp , departure_stamp , stamp), yolov11n model, 8 versions	Correct schema, far too small. Starting weights only — augment heavily with synthetic stamps
detect-for-id/passport-id	"Passport ID detector"	9,892 images , classes: glasses , sunglasses , cloth-mask , earphone , headwear	Not a passport detector . It is an ICAO photo-compliance model. Genuinely useful for checking whether a submitted photo meets passport standards — but not for document detection
document-forgery-detection/document-forgery-detection	Popular (10k views, 312 downloads)	402 images, single class name is an export artifact string	Popularity ≠ quality . Do not build your tampering claim on this
docdetection-qgqkm/pan-document-classification	real_pan / fake_pan	2,979 images, no provenance	Tempting class names, unauditable definition. Avoid
swaroopai/passport-data-prediction-jiczw	Passport fields	Not verified	Check before committing
tampering-detection/tampering-detection-1	Multi-doc tampering	Could not confirm this slug; tampering-detection-0muly exists in the same workspace (900 imgs)	Verify which you mean

Verification procedure for any Universe link (30 seconds):

Image count — under ~150 means schema reference only

Class names — integers ([0](#), [1](#), [2](#)) or export boilerplate mean unusable without manual mapping

Thumbnail — does it show a real person's card? If yes, see 7.4

7.3 Recommended Roboflow projects

Indian documents

Purpose	Project	Size	Notes
Aadhaar — best annotation	sujeet-kumar-5vze5/aadhaar-details-detection	224	15 properly named classes: aadhaar number , address , dob , photo , emblem , father , gender , goi symbol , issue date , logo , name , uiiai icon , uiiai symbol , vid , yob . Annotates <i>security elements</i> , not just text
Aadhaar — largest structured	monika-ztd2k/aadhaar-card-1-ebbdz	938	18 classes: a_qr , a_photo , a_emb , a_num , a_masked , a_govt , a_uidai , a_vid , plus card_passport , card_voter_id , p_front , p_num
Aadhaar security features	project-epimx/id-bdbwr	93	Emblem logo , Goi logo , Goi symbol , Text layer , fake . Tiny but the only one annotating security elements with a fake class
Aadhaar tampering	tampering-detection/tampering-detection-0muly	900	Tampered Hologram , Tampered image , Tampered logo , Tampered text , QR missing . Conceptually exactly Module 3 for Indian docs. Uneven annotation — audit first
Aadhaar back	mitesh-workspace/back-aadhaar-card	584	Single class aadhaar-father-name
Doc-type routing	mahrprojects/identity-card-classifier	220	aadhar / driver-license / pan / passport / voter
PAN fields	mitesh-workspace/pan-card-annotation	600	pan-name , pan-dob , pan-father-name , pan-no
PAN fields (alt)	smartxtract/pan-card-ygz7o	245	name , date_of_birth , father's_name , pan_number
PAN front/back	panbg/pan-vngzj	1,388	Largest PAN set
Aadhaar + PAN	aadharpan-yolo/aadhaar-pan-document-detection	150	MIT licensed
Driving licence	jaspreetsingh/indian-driving-licence-reader-rlxel	40	name , dl_number , dob . Schema only
Voter ID	fil-9zpqb/voter-01	146	Has trained model

Purpose	Project	Size	Notes
Passport VIZ fields	misha-88lag/passport-exe8g	328	14 field classes: <code>date_of_birth</code> , <code>expiry_date</code> , <code>doc_id</code> , <code>nationality</code> , <code>issue_date</code> , <code>place_of_birth</code> , <code>authority</code> , <code>gender</code> , etc. Highest-value model on this list — this is your VIZ↔MRZ cross-check localiser
South Asian passport fields	at-in/bangladeshi-passport-fields	200	33 classes including <code>ghost_portrait</code> , <code>holder_signature</code> , <code>issuing_authority</code> , <code>emergency_contact_*</code> . Ghost portrait is a security feature Indian passports also carry
Passport fields + face	passportdetection-rgnih/passport-fields-fio-photo	181	Includes a <code>face</code> class. License: BY-NC-SA 4.0 — flag it
Passport field classes	arvind-kumar-wjygd/passport-ppwp8	25	13 classes incl. <code>Paasport_no</code> , <code>DOE</code> , <code>DOI</code> , <code>Place_of_Issue</code> . Too small, good schema
MRZ localisation	achreffaty/mrz-ye7hu	496	Largest clean MRZ set. Classes <code>MRZ</code> , <code>MRZ_P</code>
MRZ + page	phiphi-20ww6/passport-page-mrz-detection	279	Gives <code>mrz</code> AND <code>passport_page</code> — both crops in one pass
MRZ (with model)	spacex-hken9/mrz-h6s9s	248	Has a trained model
Doc crop / deskew	ip2-kbjz5/identity-card-segmentation	157	yolov11n-seg, trained model. Segmentation gives corners for perspective correction
Doc type classify	2bs/identity-type	483	<code>Chip_Id_Card</code> , <code>UnChip_Id_Card</code> , <code>passport</code> . Has model
Stamps (largest)	shujing-liang/stamp-detection-f3yka	2,342	Trained model. Cleanest of the stamp sets
Stamps (alt)	social-code-llc/stamp-seal-detection	306	Trained model
Stamp binary	class-cemy0/stamp-wnh42	2,292	<code>Stamp</code> / <code>Non_Stamp</code>
Anti-spoofing	face-anti-spoofing-detection/face-anti-spoofing-detection	6,129	Trained model, <code>fake</code> / <code>real</code>
Anti-spoofing (classification)	project-uqhrw/face-anti-spoofing-icbck	8,269	<code>real</code> / <code>spoof</code> , has model

Purpose	Project	Size	Notes
Spoof attack types	<code>thanh-huy-xagag/anti-spoofing-avfew</code>	1,246	4 classes: <code>live</code> , <code>photo</code> , <code>device</code> , <code>mask</code> — attack-type granularity
Photo compliance	<code>detect-for-id/passport-id</code>	9,892	<code>glasses</code> , <code>sunglasses</code> , <code>cloth-mask</code> , <code>earphone</code> , <code>headwear</code> . Public Domain. Use for ICAO photo-standard checking

7.4 Provenance warning

Audit every Indian ID dataset before use. Most Aadhaar and PAN sets on Universe were built by scraping real cards off the web — you will see real names, real numbers, real faces.

If that is what you find, two options:

Use the annotation **schema only** and re-annotate your own synthetic data against it

Mask every number and face before training

Do not ship sample images from these sets in your presentation deck. An MHA panel is exactly the audience that will ask where your training data came from.

7.5 Licensing

Most Universe projects are CC BY 4.0 (attribution required). Exceptions noted above: MIT (`aadharpan-yolo`), Public Domain (`vexil-infotech`, `detect-for-id`), BY-NC-SA 4.0 (`passportdetection-rgnih`).

Include one slide listing provenance and license per dataset. Costs five minutes, preempts the most awkward question you will get.

8. Synthetic Data Generation Plan

This is roughly 40% of your data work. Treat it as a first-class deliverable — it produces both your training data *and* your live demo.

Deliverable	How	Target
Indian document templates	Vectorise official specimens (section 6.7) into layered templates: background guilloche, static text, field slots, photo slot, QR slot	5 doc types
Synthetic identities	Faker with <code>en_IN</code> locale for names/addresses. Generate Aadhaar numbers passing the Verhoeff checksum , PAN numbers matching official format rules, passport numbers matching the Indian pattern, MRZ lines with correct identities check digits	2,000
Synthetic faces	Reuse MIDV-2020's — already artificially generated, zero PII exposure. Or generate fresh with a GAN	2,000
Real Aadhaar QRs (validation only)	Team members generate their <i>own</i> offline eKYC / Secure QR from <code>myaadhaar.uidai.gov.in</code> , validate the parser, then delete. Never commit	5–10, consent only
Capture realism	Print the generated templates, then scan and photograph under varied lighting and angle. A model trained purely on clean digital renders falls apart on real scans	3 captures each
Doc-photo ↔ live-face pairs	Print a photo onto a mock ID template, scan it, capture live webcam frames. Include deliberate impostor pairs. This is what calibrates your face threshold — without it, the threshold is a guess	30–50 people, ~500 pairs
Forgeries with ground truth	Photo swap, DOB splice, MRZ↔VIZ desync, QR removal/replacement, hologram absence, copy-move. Log a ground-truth mask on every one	2,000–3,000
Passport stamps	No public dataset exists. Render synthetic stamps (circle/rectangle variants, dates, port codes) onto MIDV passport pages, then forge by duplicating and date-editing	1,000
Cross-document pairs	Matched passport+visa sets, plus deliberately inconsistent ones (name mismatch, visa outliving passport). Trivial to script; directly demos Validation Layer D	300
Watchlist seed	OFAC/UN lists + synthetic entries matching your test documents so hits actually fire in the demo	—

The print-and-rescan step is the one teams skip and should not. Two afternoons with a printer and a phone camera is the difference between a model that works on your laptop and one that works at the demo table.

9. Technology Stack

Layer	Choice	Rationale
OCR	PaddleOCR or docTR	Better on ID-card layouts than Tesseract; both have ONNX exports
MRZ parsing	Custom, ~200 lines	ICAO 9303 check-digit logic. Do not use an unmaintained wrapper
Aadhaar QR	Open-source Secure QR parser + UIDAI public certificate	Cryptographic verification, fully offline
Face	InsightFace <code>buffalo_l</code> (RetinaFace + ArcFace)	512-d embeddings, single-pass detect+align+embed
Face gallery (1:N)	Postgres <code>pgvector</code> , HNSW index	Demo-scale sufficient; mention FAISS/Milvus for production
Tampering	OpenCV (ELA, copy-move, noise residual) + PyTorch segmentation	Ensemble, not one model
Metadata	<code>exiftool</code> , <code>pikepdf</code> , Pillow	EXIF, PDF incremental-update history
Model runtime	ONNX Runtime, int8 quantised	CPU-only, warm sessions, tuned thread count
Backend	FastAPI + WebSocket	Async fan-out, streaming results
Store	Postgres + MinIO	Postgres for cases/audit, MinIO for images
Frontend	React + Vite	Officer console
Packaging	Docker Compose, fully offline	Border posts have no reliable internet
Training/annotation	Roboflow (build-time only)	Export weights, run locally

10. Repository Structure




```
├── voter_id.yaml
├── driving_licence.yaml
├── models/          # ONNX weights, baked into image
├── data/
│   ├── generator/   # synthetic doc + forgery generator
│   ├── templates/   # vectorised specimen templates
│   └── ground_truth/ # masks, labels
├── ui/              # React officer console
├── docker-compose.yml
└── tests/
```

11. Development Roadmap

Phase 0 — Week 0 (this week)

Freeze the `Signal` contract and profile schema. Everything else depends on it.

Stand up FastAPI with a stub endpoint returning fake signals

Build the React console against those stubs

Download MIDV-2020

Everyone now has a target to code against, and the UI is never blocked.

Phase 1 — Weeks 1-2: Extraction & validation spine

Fork `ip2-kbjz5/identity-card-segmentation` and `achreffaty/mrz-ye7hu`, export ONNX

Wire: crop → MRZ detect → PaddleOCR (charset-restricted) → parser → check digits

Implement validation Layers A-C

Implement the Aadhaar Secure QR verifier

Deliverable: upload a passport or Aadhaar, get a validated field set with per-field checksum/signature status.

This alone is a demoable system.

Phase 2 — Weeks 3-4: Cross-checks & face

Train the VIZ field detector from `misha-88lag/passport-exe8g`

Add the VIZ↔MRZ mismatch signal

Implement Layer D across passport + visa

InsightFace 1:1 verify, threshold calibrated on doc-vs-live pairs

pgvector gallery for 1:N duplicate identity

Deliverable: cross-document contradictions surfaced, face verdict, duplicate-identity flag.

Phase 3 — Weeks 5–6: Tampering

Classical signals first: ELA, copy-move, noise residual, stamp duplicates, metadata

Build the synthetic forgery generator

Annotate in Roboflow, train a segmentation model for the heatmap

Physical-track signals: OCR-B conformance, layout geometry, guilloche

Deliverable: tamper heatmap with region-level evidence.

Phase 4 — Weeks 7–8: Hardening & story

Anti-spoofing model integrated

FAR/FRR curve exposed in the UI

Bias evaluation across FairFace/RFW, results documented

Audit log complete

Offline Docker Compose install verified with the network cable pulled

Rehearse the live-tampering demo **and one honest failure case**

Team split (six people)

Role	Scope
2 × Extraction + Validation	Largest surface area
1 × Face + Gallery	1:1, 1:N, liveness, calibration
2 × Tampering + Data generation	Both tracks + synthetic generator
1 × API + Fusion + Console	Integration owner — your strongest generalist

Cut list (if behind schedule)

Cut in this order:

Active liveness (keep passive only)

Learned tamper segmentation (classical signals alone are defensible)

Layer F temporal analysis

Driving licence and permit support (passport + visa + Aadhaar proves the architecture generalises)

Do not cut:

MRZ check digits

VIZ↔MRZ cross-check

Aadhaar QR signature verification

Layer D cross-document validation

The evidence list in the UI

These five are what make the system credible to an SSB officer, and four of them are nearly free.

12. Legal & Compliance

Aadhaar

Aadhaar Act 2016, Section 29 restricts sharing of Aadhaar information. UIDAI explicitly prohibits service providers from sharing, publishing, or displaying the offline eKYC XML or its contents. Non-compliance invites action under Sections 29(2), 29(3), 29(4) and 37 of the Act.

Support masked Aadhaar (last four digits only)

Store a salted hash, never the number

Verify and discard — do not retain

DPDP Act 2023

Data minimisation: collect only what the screening decision requires

Purpose limitation: screening data is not repurposed

Storage limitation: define and enforce retention periods

Encrypt at rest and in transit

Face data

Store **embeddings + audit logs, not raw face images**

Do not retain live captures beyond the session unless legally mandated

Encrypt the embedding store

Bias testing is a compliance item, not a nice-to-have. NIST FRVT documents accuracy disparities across demographics; in a border-security context those disparities have real consequences

Training data provenance

Every source used must be synthetic, official specimen, or officially published

Do not use scraped Indian ID datasets. Several circulate on Kaggle and GitHub with names like "Aadhaar card images" or "Indian ID OCR dataset". Many contain real people's documents uploaded without consent

Maintain a provenance + license table and put it on a slide

System positioning

Human-in-the-loop. The system assists, never replaces, the officer's decision

Full audit trail: score, model version, timestamp, officer ID, per-signal evidence

Reproducibility: a decision must be reconstructable months later for an investigation

13. Resource Links Appendix

Core datasets

Resource	Link
MIDV-2020 (FTP)	<code>ftp://smartengines.com/midv-2020</code>
MIDV-2020 (mirror + docs)	https://l3i-share.univ-lr.fr/MIDV2020/midv2020.html
MIDV-2020 paper	https://arxiv.org/abs/2107.00396
MIDV-500	<code>ftp://smartengines.com/midv-500</code>
MIDV-2019	<code>ftp://smartengines.com/midv-2019</code>
MIDV-Holo	https://github.com/SmartEngines/midv-holo
MIDV-Holo data	<code>ftp://smartengines.com/midv-holo</code>
SIDTD	https://github.com/Oriolrt/SIDTD_Dataset
SIDTD (CVC repository)	https://tc11.cvc.uab.es/datasets/SIDTD_1/
SIDTD paper	https://www.nature.com/articles/s41597-024-04160-9
IDNet paper	https://arxiv.org/abs/2408.01690
DocXPand-25k	https://github.com/quicksign/docxpannd
DocTamper	https://github.com/qcf-568/DocTamper
MIDV-DynAttack	https://github.com/EPITAResearchLab/pouliquen.25.icdar
CelebA-Spoof	https://github.com/ZhangYuanhan-AI/CelebA-Spoof
FairFace	https://github.com/joojs/fairface
CoMoFoD	https://www.vcl.fer.hr/comofod/

Official / government

Resource	Link
UIDAI QR Code Reader + public certificate	https://uidai.gov.in/en/ecosystem/authentication-devices-documents/qr-code-reader.html

Resource	Link
UIDAI documentation	https://docs.uidai.gov.in/
UIDAI offline eKYC (resident portal)	https://myaadhaar.uidai.gov.in/offline-ekyc
PRADO — authentic document register	https://www.consilium.europa.eu/prado/
ICAO Doc 9303	https://www.icao.int/publications/pages/publication.aspx?docnum=9303
OFAC SDN sanctions list	https://ofac.treasury.gov/sanctions-list-service
UN Security Council Consolidated List	https://scsanctions.un.org/

Implementation references

Resource	Link
Roboflow Inference SDK	https://inference.roboflow.com/inference_helpers/inference_sdk/
Aadhaar Offline KYC (Android reference)	https://github.com/GovindaPaliwal/Aadhaar-Offline-KYC-Android-Library
InsightFace	https://github.com/deepinsight/insightface
PaddleOCR	https://github.com/PaddlePaddle/PaddleOCR
docTR	https://github.com/mindee/doctr
pgvector	https://github.com/pgvector/pgvector
ONNX Runtime	https://onnxruntime.ai/

Roboflow Universe — recommended projects

Indian documents

<https://universe.roboflow.com/sujeet-kumar-5vze5/aadhaar-details-detection>
<https://universe.roboflow.com/monika-ztd2k/aadhaar-card-1-ebbdz>
<https://universe.roboflow.com/project-epimx/id-bdbwr>
<https://universe.roboflow.com/tampering-detection/tampering-detection-0muly>
<https://universe.roboflow.com/mitesh-workspace/back-aadhaar-card>

<https://universe.roboflow.com/mahrprojects/identity-card-classifier>
<https://universe.roboflow.com/mitesh-workspace/pan-card-annotation>
<https://universe.roboflow.com/smartxtract/pan-card-ygz7o>
<https://universe.roboflow.com/panbg/pan-vngzj>
<https://universe.roboflow.com/aadharpan-yolo/aadhaar-pan-document-detection>
<https://universe.roboflow.com/jaspreetsingh/indian-driving-licence-reader-rlxel>
<https://universe.roboflow.com/fil-9zpgb/voter-01>

International / generic

<https://universe.roboflow.com/misha-88lag/passport-exe8g>
<https://universe.roboflow.com/at-in/bangladeshi-passport-fields>
<https://universe.roboflow.com/achreffaty/mrz-ye7hu>
<https://universe.roboflow.com/phiphi-20ww6/passport-page-mrz-detection>
<https://universe.roboflow.com/spacex-hken9/mrz-h6s9s>
<https://universe.roboflow.com/ip2-kbjz5/identity-card-segmentation>
<https://universe.roboflow.com/2bs/identity-type>
<https://universe.roboflow.com/shujing-liang/stamp-detection-f3yka>
<https://universe.roboflow.com/social-code-llc/stamp-seal-detection>
<https://universe.roboflow.com/passport-stamp/passport-stamp>
<https://universe.roboflow.com/face-anti-spoofing-detection/face-anti-spoofing-detection>
<https://universe.roboflow.com/project-uqhrw/face-anti-spoofing-icbck>
<https://universe.roboflow.com/thanh-huy-xagag/anti-spoofing-avfew>
<https://universe.roboflow.com/detect-for-id/passport-id>

Verified as problematic — avoid or handle with care

`document-forgery-detection/document-forgery-detection` — 402 imgs, export-artifact class name
`docdetection-qgqkm/pan-document-classification` — unauditable `real_pan`/`fake_pan`
`centurion-.../aadhaar-card-details-extraction-n4hyw` and `cutm-iwh4a/aadhaar-card-details` — same dataset, unnamed integer classes
`vexil-infotech/voterid-card-details-extraction` — 31 images, schema only

Document compiled for SIH PS 26188. Verify all Roboflow image counts and class names before committing — Universe projects change without notice.